



Nguyen Viet Hoang

SUMMARY

*AI Engineer (1+ yr) specializing in NLP, embeddings, and retrieval-augmented systems — building agentic AI, multi-agent workflows (LangGraph), and information retrieval pipelines on async Python/FastAPI.
Core: NLP & Embeddings · Information Retrieval (RAG) · Agentic AI (LangGraph) · Backend (FastAPI, Supabase/pgvector) · Workflow Automation (n8n)*

Phone:

0383700402

Email:

ngviethoang.work@gmail.com

Address:

Ho Chi Minh City, Vietnam

EXPERIENCE

Founding Engineer – Independent Startup (03/2026 – Present)

Built and delivered a document OCR solution for an external company, digitizing paperwork-based workflows into structured data.

Led the technical build end-to-end, using CLI-based agentic tools like Claude Code to accelerate delivery.

AI Engineer - Mebisoft (08/2025 – 5/2026)

-Agentic Text-to-SQL Assistant (RAG + Guardrails): Built a chatbot letting farmers query a 62-table database in natural language (Vietnamese) using retrieval-augmented generation with guardrails to reduce hallucination – >90% accuracy.

-Performance Optimization: Reduced inference latency from 8s to 1.5s per image (81% improvement) while handling 2,000+ daily requests, lowering compute cost per request.

-Shrimp Disease Diagnosis via Image Analysis: Built a classification system to detect common shrimp diseases from images, combined with historical/environmental data for early warnings – >95% accuracy.

-Automated Shrimp Length Measurement: Built a CV system (YOLO/RTMDet) to detect and measure shrimp length, replacing manual measurement – >95% accuracy.

-CI/CD & Monitoring: Set up GitHub Actions for automated testing/deployment to Azure; implemented Azure Monitor and Application Insights for real-time monitoring – >99% uptime.

AI Engineer Intern – ThinkWare AI Solutions (01/2025 – 06/2025)

RAG & Agent Evaluation: Designed evaluation pipelines (Hit Rate, MRR) to score prompts and grade agent responses, improving response accuracy by 25%.

Data Curation: Curated instruction-tuning datasets, boosting tool-calling reliability by 20%.

Annotation: Labeled 10,000+ objects for autonomous driving with high precision.

Automation: Built Python scripts reducing manual effort by 40%.

Python Instructor (08/2024 – 12/2024):

Taught Python to 10+ high school students; built a web scraping automation pipeline reducing manual effort by 40%.

PROJECTS

VietStoryLens - AI Landmark Recognition & Storytelling

LotusHacks x HackHarvard 2026

Built an end-to-end AI app recognizing 15 HCMC landmarks from photos (CLIP + FAISS, 85-93% accuracy) via a LangGraph Router-Verifier RAG pipeline (DeepSeek V4) with zero hallucination, exposed through an MCP server.

Key highlight: Demoted to NVIDIA's Software Director at the expo — recognized for the agentic RAG architecture, leading to direct engagement with NVIDIA's AI community.

Tech stack: CLIP, FAISS, LangGraph, DeepSeek V4, FastAPI, Docker, MCP.

Slack AI Assistant (n8n + Supabase pgvector):

RAG bot with hybrid retrieval, tool-calling & thread-aware memory, answering team questions in Slack.

AI-Powered Fungi Detection System

Leader-Olympic AI HCMC Competition

- Designed a hybrid ConvNeXt + Transformer model, achieving 97.7% accuracy in fungal species classification.
- Automated the end-to-end pipeline: data ingestion, cleaning, augmentation, and preprocessing.
- Led a 4-member team, coordinated research direction, and prepared presentation materials.

HONORS & AWARDS

4th | Olympic AI HCMC Competition 2025

Consolation Prize | School IQ Math Competition

Contributor – Hermes Desktop (13.2k★) – Fixed rate-limiting bug affecting thousands of users.

Outstanding Graduate of the Class of 2023

Champion of PieoneDream Hackathon AI x Blockchain

EDUCATION

Bachelor of Machine Learning | 2022-2026

HCMC University of Technology

CERTIFICATES

- **Google** AI essentials
- **Google** Data Analytics
- **Stanford** University Machine Learning Specialization
- **Stanford** AI in Healthcare
- **IBM** AI Engineering Professional Certificate

ACTIVITIES

- Facilitator Of DevFest - Google Developer Group
 - Facilitator của QuanQuanGCP25 - Google Developer Group
 - Research Assistant | AI Scientific Research Laboratory
 - Vietnam National Student Olympiad 2025 Participant
 - Youth Digital Citizens Challenge **Top 40/200**
 - HumanLog Competition by Kühne Logistics University **Top 60 /1000**
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